PRACTICAL INSTRUMENTATION REFERENCE

OpenTelemetry GenAI Attribute Checklist

Compiled from the OpenTelemetry GenAI semantic conventions

SOURCE REPOSITORIES

https://github.com/open-telemetry/semantic-conventions-genai
https://github.com/open-telemetry/semantic-conventions

PINNED COMMITS

`c88d504ab3d9879f8e50d3cc87e69775e11db234`
`d0472f4ae331e8ef01aa571fe024d0d6a1a9b5e1`

COMPILED

2026-09-21

REGISTRY SNAPSHOT

72
`gen_ai.*`
attributes

Unofficial compilation

Not an OpenTelemetry publication.

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READ THIS FIRST

How to use this checklist

01 Choose the signal and operation that match the boundary you are instrumenting.

02 Apply the published span name and span kind where the source defines them.

03 Work down the rows in requirement order. Keep each source condition attached to its row.

04 Treat content-bearing opt-in fields carefully; the source warns that several may contain sensitive information.

05 Use Appendix A to look up every registry entry, including attributes not assigned to one checklist here.

Requirement legend

REQUIRED	Published as required for that signal.
CONDITIONALLY REQUIRED	Required when the source condition applies.
RECOMMENDED	Published as recommended, including any row condition.
OPT-IN	Published as opt-in rather than captured by default.

Stability legend

STABLE	Stable in the pinned source.
RELEASE CANDIDATE	Release Candidate in the pinned source.
DEVELOPMENT	Development in the pinned source.

On opentelemetry.io, every current GenAI attribute carries a Deprecated badge only because the conventions moved to the new repository. This checklist follows that new GenAI semantic conventions repository; the separate table below lists the ten names that are actually deprecated.

Deprecated - do not use

OLD REGISTRY SOURCE ↗

ATTRIBUTE	WHAT TO USE INSTEAD	NOTE
gen_ai.completion	Deprecated, use Event API to report completions contents.	Removed, no replacement at this time.
gen_ai.prompt	Deprecated, use Event API to report prompt contents.	Removed, no replacement at this time.
gen_ai.system	Deprecated, use gen_ai.provider.name instead.	Replaced by gen_ai.provider.name, which has moved to the OpenTelemetry GenAI semantic conventions repository.
gen_ai.usage.completion_tokens	Deprecated, use gen_ai.usage.output_tokens instead.	Replaced by gen_ai.usage.output_tokens, which has moved to the OpenTelemetry GenAI semantic conventions repository.
gen_ai.usage.prompt_tokens	Deprecated, use gen_ai.usage.input_tokens instead.	Replaced by gen_ai.usage.input_tokens, which has moved to the OpenTelemetry GenAI semantic conventions repository.
gen_ai.openai.request.response_format	Deprecated, use gen_ai.output.type.	Replaced by gen_ai.output.type, which has moved to the OpenTelemetry GenAI semantic conventions repository.
gen_ai.openai.request.seed	Deprecated, use gen_ai.request.seed.	Replaced by gen_ai.request.seed, which has moved to the OpenTelemetry GenAI semantic conventions repository.
gen_ai.openai.request.service_tier	Deprecated, use openai.request.service_tier.	Replaced by openai.request.service_tier, which has moved to the OpenTelemetry GenAI semantic conventions repository.
gen_ai.openai.response.service_tier	Deprecated, use openai.response.service_tier.	Replaced by openai.response.service_tier, which has moved to the OpenTelemetry GenAI semantic conventions repository.
gen_ai.openai.response.system_fingerprint	Deprecated, use openai.response.system_fingerprint.	Replaced by openai.response.system_fingerprint, which has moved to the OpenTelemetry GenAI semantic conventions repository.

Scope. Attribute names, types, requirement levels, stability, descriptions, examples, metric metadata, and span conventions are compiled only from the pinned repository files. Descriptions and examples are exact source text or shortened exact prefixes.

CORE CLIENT TELEMETRY

LLM inference span

Apply the published inference span convention to model or service calls that generate a response or request a tool call.

Inference

SOURCE ↗

SPAN NAME

SHOULD be {gen_ai.operation.name} {gen_ai.request.model}. Semantic conventions for individual GenAI systems and frameworks MAY specify different span name format and MUST follow the overall guidelines for span names.

SPAN KIND

SHOULD be CLIENT and MAY be set to INTERNAL on spans representing call to models running in the same process. It's RECOMMENDED to use CLIENT kind when the GenAI system being instrumented usually runs in a different process than its client or when the GenAI call happens over instrumented protocol such as HTTP.

ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
gen_ai.operation.name DEVELOPMENT	string	REQUIRED	The name of the operation being performed.	chat; generate_content; text_completion
gen_ai.provider.name DEVELOPMENT	string	REQUIRED	The Generative AI provider as identified by the client or server instrumentation.	openai; gcp.gen_ai; gcp.vertex_ai
error.type STABLE	string	CONDITIONALLY REQUIRED If the operation ended in an error	Describes a class of error the operation ended with.	timeout; java.net.UnknownHostException; server_certificate_invalid; 500
gen_ai.conversation.id DEVELOPMENT	string	CONDITIONALLY REQUIRED	The unique identifier for a conversation (session, thread), used to store and correlate messages within this conversation.	conv_5j66UpCpwteGg4YSxUnt7lPY
gen_ai.output.type DEVELOPMENT	string	CONDITIONALLY REQUIRED	Represents the content type requested by the client.	text; json; image
gen_ai.prompt.name DEVELOPMENT	string	CONDITIONALLY REQUIRED when a named prompt template is used	The name of the prompt that uniquely identifies it.	analyze-code
gen_ai.prompt.version DEVELOPMENT	string	CONDITIONALLY REQUIRED	The version of the prompt template used.	1.0.0; 2025-05-01; prod; v2
gen_ai.request.choice.count DEVELOPMENT	int	CONDITIONALLY REQUIRED If available, in the request, and !=1	The target number of candidate completions to return.	3

☐	ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
☐	**gen_ai.request.model** DEVELOPMENT	string	CONDITIONALLY REQUIRED If available	The name of the GenAI model a request is being made to.	gpt-4
☐	**gen_ai.request.seed** DEVELOPMENT	int	CONDITIONALLY REQUIRED If applicable and if the request includes a seed	Requests with same seed value more likely to return same result.	100
☐	**gen_ai.request.stream** DEVELOPMENT	boolean	CONDITIONALLY REQUIRED	Indicates whether the GenAI request was made in streaming mode.	–
☐	**gen_ai.request.top_k** DEVELOPMENT	int	CONDITIONALLY REQUIRED If applicable	The top-K sampling setting for the GenAI request: restricts token generation at each step to the K most likely next tokens.	40
☐	**server.port** STABLE	int	CONDITIONALLY REQUIRED If server.address is set	GenAI server port.	80; 8080; 443
☐	**gen_ai.conversation.compacted** DEVELOPMENT	boolean	RECOMMENDED when available	Indicates whether the effective conversation context used for this operation is a compacted view of a prior conversation.	true
☐	**gen_ai.request.frequency_penalty** DEVELOPMENT	double	RECOMMENDED	The frequency penalty setting for the GenAI request.	0.1
☐	**gen_ai.request.max_tokens** DEVELOPMENT	int	RECOMMENDED	The maximum number of tokens the model generates for a request.	100
☐	**gen_ai.request.presence_penalty** DEVELOPMENT	double	RECOMMENDED	The presence penalty setting for the GenAI request.	0.1
☐	**gen_ai.request.previous_response.id** DEVELOPMENT	string	RECOMMENDED	The unique identifier of a previous response or interaction used to provide context for the current operation.	resp_0123456789aBCdef; interaction-123
☐	**gen_ai.request.reasoning.level** DEVELOPMENT	string	RECOMMENDED When applicable	The reasoning or thinking effort level requested for a GenAI model.	low; medium; high
☐	**gen_ai.request.stop_sequences** DEVELOPMENT	string[]	RECOMMENDED	List of sequences that the model will use to stop generating further tokens.	["forest", "lived"]
☐	**gen_ai.request.temperature** DEVELOPMENT	double	RECOMMENDED	The temperature setting for the GenAI request.	0.0

☐	ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
☐	**gen_ai.request.top_p** DEVELOPMENT	double	RECOMMENDED	The top_p sampling setting for the GenAI request.	1.0
☐	**gen_ai.response.finish_reasons** DEVELOPMENT	string[]	RECOMMENDED	Array of reasons the model stopped generating tokens, corresponding to each generation received.	["stop"]; ["stop", "length"]; ["stop", "length", "error"]
☐	**gen_ai.response.id** DEVELOPMENT	string	RECOMMENDED	The unique identifier for the completion.	chatcmpl-123
☐	**gen_ai.response.model** DEVELOPMENT	string	RECOMMENDED	The name of the model that generated the response.	gpt-4-0613
☐	**gen_ai.response.time_to_first_chunk** DEVELOPMENT	double	RECOMMENDED If the request was a streaming request	Time to first chunk in a streaming response, measured from request issuance, in seconds. The value is measured from when the client issues the generation request to when…	0.5; 1.2
☐	**gen_ai.usage.audio.cache_read.input_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of audio input tokens served from a provider-managed cache.	60
☐	**gen_ai.usage.audio.input_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of audio input tokens.	120
☐	**gen_ai.usage.audio.output_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of audio output tokens.	240
☐	**gen_ai.usage.cache_read.input_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of input tokens served from a provider-managed cache.	50
☐	**gen_ai.usage.cache_write.input_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of input tokens written to a provider-managed cache.	25
☐	**gen_ai.usage.image.cache_read.input_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of image input tokens served from a provider-managed cache.	128
☐	**gen_ai.usage.image.input_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of image input tokens.	258

☐	ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
☐	**gen_ai.usage.image.output_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of image output tokens.	1290
☐	**gen_ai.usage.input_tokens** DEVELOPMENT	int	RECOMMENDED	The number of tokens used in the GenAI input (prompt).	100
☐	**gen_ai.usage.output_tokens** DEVELOPMENT	int	RECOMMENDED	The number of tokens used in the GenAI response (completion).	180
☐	**gen_ai.usage.reasoning.output_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of output tokens used for reasoning (e.g. chain-of-thought, extended thinking).	50
☐	**gen_ai.usage.text.cache_read.input_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of text input tokens served from a provider-managed cache.	40
☐	**gen_ai.usage.text.input_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of text input tokens.	100
☐	**gen_ai.usage.text.output_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of text output tokens.	180
☐	**server.address** STABLE	string	RECOMMENDED	GenAI server address.	example.com; 10.1.2.80; /tmp/my.sock
☐	**gen_ai.input.messages** DEVELOPMENT	any	OPT-IN	The chat history provided to the model as an input.	[{ "role": "user", "parts": [{ "type": "text", "content": "Weather in Paris?" }] }, {…
☐	**gen_ai.output.messages** DEVELOPMENT	any	OPT-IN	Messages returned by the model where each message represents a specific model response (choice, candidate).	[{ "role": "assistant", "parts": [{ "type": "text", "content": "The weather in Paris is…
☐	**gen_ai.prompt.variable** DEVELOPMENT	string	OPT-IN	The variables supplied to the prompt template, the being the variable name, the value being the variable value.	Alice; French
☐	**gen_ai.system_instructions** DEVELOPMENT	any	OPT-IN	The system message or instructions provided to the GenAI model separately from the chat history.	[{ "type": "text", "content": "You are an Agent that greet users, always use greetings…

☐	ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
☐	**gen_ai.tool.definitions** DEVELOPMENT	any	OPT-IN	The list of tool definitions available to the GenAI agent or model.	`[ { "type": "function", "name": "get_current_weather", "description": "Get the current…`

VECTOR GENERATION

Embeddings span

Use this convention for requests that generate embeddings from an input.

Embeddings

SOURCE ↗

SPAN NAME
SHOULD be {gen_ai.operation.name} {gen_ai.request.model}.

SPAN KIND
SHOULD be CLIENT.

☐	ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
☐	**gen_ai.operation.name** DEVELOPMENT	string	REQUIRED	The name of the operation being performed.	chat; generate_content; text_completion
☐	**gen_ai.provider.name** DEVELOPMENT	string	REQUIRED	The Generative AI provider as identified by the client or server instrumentation.	openai; gcp.gen_ai; gcp.vertex_ai
☐	**error.type** STABLE	string	CONDITIONALLY REQUIRED If the operation ended in an error	Describes a class of error the operation ended with.	timeout; java.net.UnknownHostException; server_certificate_invalid; 500
☐	**gen_ai.request.model** DEVELOPMENT	string	CONDITIONALLY REQUIRED If available	The name of the GenAI model a request is being made to.	gpt-4
☐	**server.port** STABLE	int	CONDITIONALLY REQUIRED If server.address is set	GenAI server port.	80; 8080; 443
☐	**gen_ai.embeddings.dimension.count** DEVELOPMENT	int	RECOMMENDED	The number of dimensions the resulting output embeddings should have.	512; 1024
☐	**gen_ai.request.encoding_formats** DEVELOPMENT	string[]	RECOMMENDED	The encoding formats requested in an embeddings operation, if specified.	["base64"]; ["float", "binary"]
☐	**gen_ai.response.model** DEVELOPMENT	string	RECOMMENDED	The name of the model that generated the response.	gpt-4-0613
☐	**gen_ai.usage.input_tokens** DEVELOPMENT	int	RECOMMENDED	The number of tokens used in the GenAI input (prompt).	100

☐	ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
☐	**server.address** STABLE	string	RECOMMENDED	GenAI server address.	example.com; 10.1.2.80; /tmp/my.sock

AGENT TOOL CALLS

Tool-execution span

Use this convention for tool execution, including tool calls made directly by application code.

Execute tool

SOURCE ↗

SPAN NAME
SHOULD be execute_tool {gen_ai.tool.name}.

SPAN KIND
SHOULD be INTERNAL.

ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
gen_ai.operation.name DEVELOPMENT	string	REQUIRED	The name of the operation being performed.	chat; generate_content; text_completion
gen_ai.tool.name DEVELOPMENT	string	REQUIRED	Name of the tool utilized by the agent.	Flights
error.type STABLE	string	CONDITIONALLY REQUIRED If the operation ended in an error	Describes a class of error the operation ended with.	timeout; java.net.UnknownHostException; server_certificate_invalid; 500
gen_ai.agent.name DEVELOPMENT	string	CONDITIONALLY REQUIRED When applicable	The human-readable name of the agent executing the tool.	Math Tutor; Fiction Writer
gen_ai.conversation.id DEVELOPMENT	string	CONDITIONALLY REQUIRED If available	The unique identifier for a conversation (session, thread), used to store and correlate messages within this conversation.	conv_5j66UpCpwteGg4YSxUnt7lPY
gen_ai.tool.call.id DEVELOPMENT	string	RECOMMENDED If available	The tool call identifier.	call_mszuSIzqtI65i1wAUOE8w5H4
gen_ai.tool.description DEVELOPMENT	string	RECOMMENDED If available	The tool description.	Multiply two numbers
gen_ai.tool.type DEVELOPMENT	string	RECOMMENDED If available	Type of the tool utilized by the agent	function; extension; datastore

☐	ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
☐	**gen_ai.tool.call.arguments** DEVELOPMENT	any	OPT-IN	Parameters passed to the tool call.	`{ "location": "San Francisco?", "date": "2025-10-01" }`
☐	**gen_ai.tool.call.result** DEVELOPMENT	any	OPT-IN	The result returned by the tool call (if any and if execution was successful).	`{ "temperature_range": { "high": 75, "low": 60 }, "conditions": "sunny" }`

CREATE, INVOKE, COORDINATE, PLAN

Agent and workflow spans

Choose the operation that matches the agent or framework boundary being instrumented.

Create agent span

SOURCE ↗

SPAN NAME

SHOULD be `create_agent {gen_ai.agent.name}`. Semantic conventions for individual GenAI systems and frameworks MAY specify different span name format.

SPAN KIND

SHOULD be `CLIENT`.

ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
`gen_ai.operation.name` DEVELOPMENT	string	REQUIRED	The name of the operation being performed.	`chat; generate_content; text_completion`
`gen_ai.provider.name` DEVELOPMENT	string	REQUIRED	The Generative AI provider as identified by the client or server instrumentation.	`openai; gcp.gen_ai; gcp.vertex_ai`
`error.type` STABLE	string	CONDITIONALLY REQUIRED If the operation ended in an error	Describes a class of error the operation ended with.	`timeout; java.net.UnknownHostException; server_certificate_invalid; 500`
`gen_ai.agent.description` DEVELOPMENT	string	CONDITIONALLY REQUIRED If provided by the application	The free-form description of the GenAI agent created during this operation.	`Helps with math problems; Generates fiction stories`
`gen_ai.agent.id` DEVELOPMENT	string	CONDITIONALLY REQUIRED If applicable	The stable unique identifier of the GenAI agent created during this operation.	`asst_5j66UpCpwteGg4YSxUnt7lPY; arn:aws:bedrock:us-east-1:123:agent/42…`
`gen_ai.agent.name` DEVELOPMENT	string	CONDITIONALLY REQUIRED If provided by the application	The human-readable name of the GenAI agent created during this operation.	`Math Tutor; Fiction Writer`
`gen_ai.agent.version` DEVELOPMENT	string	CONDITIONALLY REQUIRED If provided by the application	The version of the GenAI agent created during this operation.	`1.0.0; 2025-05-01`
`gen_ai.request.model` DEVELOPMENT	string	CONDITIONALLY REQUIRED If available	The name of the GenAI model a request is being made to.	`gpt-4`

ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
server.port STABLE	int	CONDITIONALLY REQUIRED If server.address is set	GenAI server port.	80; 8080; 443
server.address STABLE	string	RECOMMENDED	GenAI server address.	example.com; 10.1.2.80; /tmp/my.sock
gen_ai.system_instructions DEVELOPMENT	any	OPT-IN	The system message or instructions provided to the GenAI model separately from the chat history.	[{ "type": "text", "content": "You are an Agent that greet users, always use greetings…

Invoke agent client span

SOURCE ↗

SPAN NAME

SHOULD be invoke_agent {gen_ai.agent.name} if gen_ai.agent.name is readily available. When gen_ai.agent.name is not available, it SHOULD be invoke_agent. Semantic conventions for individual GenAI systems and frameworks MAY specify different span name format.

SPAN KIND

SHOULD be CLIENT.

ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
gen_ai.operation.name DEVELOPMENT	string	REQUIRED	The name of the operation being performed.	chat; generate_content; text_completion
gen_ai.provider.name DEVELOPMENT	string	REQUIRED	The Generative AI provider as identified by the client or server instrumentation.	openai; gcp.gen_ai; gcp.vertex_ai
error.type STABLE	string	CONDITIONALLY REQUIRED If the operation ended in an error	Describes a class of error the operation ended with.	timeout; java.net.UnknownHostException; server_certificate_invalid; 500
gen_ai.agent.description DEVELOPMENT	string	CONDITIONALLY REQUIRED When available	The free-form description of the invoked GenAI agent.	Helps with math problems; Generates fiction stories
gen_ai.agent.id DEVELOPMENT	string	CONDITIONALLY REQUIRED If applicable	The stable unique identifier of the invoked GenAI agent.	asst_5j66UpCpwteGg4YSxUnt7lPY; arn:aws:bedrock:us-east-1:123:agent/42…
gen_ai.agent.name DEVELOPMENT	string	CONDITIONALLY REQUIRED When available	The human-readable name of the invoked GenAI agent.	Math Tutor; Fiction Writer

☐	ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
☐	**gen_ai.agent.version** DEVELOPMENT	string	CONDITIONALLY REQUIRED When available	The version of the invoked GenAI agent.	1.0.0; 2025-05-01
☐	**gen_ai.conversation.id** DEVELOPMENT	string	CONDITIONALLY REQUIRED	The unique identifier for a conversation (session, thread), used to store and correlate messages within this conversation.	conv_5j66UpCpwteGg4YSxUnt7lPY
☐	**gen_ai.data_source.id** DEVELOPMENT	string	CONDITIONALLY REQUIRED If applicable	The data source identifier.	H7STPQYOND
☐	**gen_ai.output.type** DEVELOPMENT	string	CONDITIONALLY REQUIRED	Represents the content type requested by the client.	text; json; image
☐	**gen_ai.request.choice.count** DEVELOPMENT	int	CONDITIONALLY REQUIRED If available, in the request, and !=1	The target number of candidate completions to return.	3
☐	**gen_ai.request.seed** DEVELOPMENT	int	CONDITIONALLY REQUIRED If applicable and if the request includes a seed	Requests with same seed value more likely to return same result.	100
☐	**server.port** STABLE	int	CONDITIONALLY REQUIRED If server.address is set	GenAI server port.	80; 8080; 443
☐	**gen_ai.request.frequency_penalty** DEVELOPMENT	double	RECOMMENDED	The frequency penalty setting for the GenAI request.	0.1
☐	**gen_ai.request.max_tokens** DEVELOPMENT	int	RECOMMENDED	The maximum number of tokens the model generates for a request.	100
☐	**gen_ai.request.model** DEVELOPMENT	string	RECOMMENDED If applicable	The name of the GenAI model configured for the agent.	gpt-4
☐	**gen_ai.request.presence_penalty** DEVELOPMENT	double	RECOMMENDED	The presence penalty setting for the GenAI request.	0.1
☐	**gen_ai.request.previous_response.id** DEVELOPMENT	string	RECOMMENDED	The unique identifier of a previous response or interaction used to provide context for the current operation.	resp_0123456789aBCdef; interaction-123
☐	**gen_ai.request.stop_sequences** DEVELOPMENT	string[]	RECOMMENDED	List of sequences that the model will use to stop generating further tokens.	["forest", "lived"]

☐	ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
☐	**gen_ai.request.temperature** DEVELOPMENT	double	RECOMMENDED	The temperature setting for the GenAI request.	0.0
☐	**gen_ai.request.top_p** DEVELOPMENT	double	RECOMMENDED	The top_p sampling setting for the GenAI request.	1.0
☐	**gen_ai.response.finish_reasons** DEVELOPMENT	string[]	RECOMMENDED	Array of reasons the model stopped generating tokens, corresponding to each generation received.	["stop"]; ["stop", "length"]; ["stop", "length", "error"]
☐	**gen_ai.usage.audio.cache_read.input_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of audio input tokens served from a provider-managed cache.	60
☐	**gen_ai.usage.audio.input_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of audio input tokens.	120
☐	**gen_ai.usage.audio.output_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of audio output tokens.	240
☐	**gen_ai.usage.cache_read.input_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of input tokens served from a provider-managed cache.	50
☐	**gen_ai.usage.cache_write.input_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of input tokens written to a provider-managed cache.	25
☐	**gen_ai.usage.image.cache_read.input_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of image input tokens served from a provider-managed cache.	128
☐	**gen_ai.usage.image.input_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of image input tokens.	258
☐	**gen_ai.usage.image.output_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of image output tokens.	1290
☐	**gen_ai.usage.input_tokens** DEVELOPMENT	int	RECOMMENDED	The number of tokens used in the GenAI input (prompt).	100
☐	**gen_ai.usage.output_tokens** DEVELOPMENT	int	RECOMMENDED	The number of tokens used in the GenAI response (completion).	180

ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
gen_ai.usage.text.cache_read.input_tokens DEVELOPMENT	int	RECOMMENDED When applicable	The number of text input tokens served from a provider-managed cache.	40
gen_ai.usage.text.input_tokens DEVELOPMENT	int	RECOMMENDED When applicable	The number of text input tokens.	100
gen_ai.usage.text.output_tokens DEVELOPMENT	int	RECOMMENDED When applicable	The number of text output tokens.	180
server.address STABLE	string	RECOMMENDED	GenAI server address.	example.com; 10.1.2.80; /tmp/my.sock
gen_ai.input.messages DEVELOPMENT	any	OPT-IN	The chat history provided to the model as an input.	[{ "role": "user", "parts": [{ "type": "text", "content": "Weather in Paris?" }] }, {…
gen_ai.output.messages DEVELOPMENT	any	OPT-IN	Messages returned by the model where each message represents a specific model response (choice, candidate).	[{ "role": "assistant", "parts": [{ "type": "text", "content": "The weather in Paris is…
gen_ai.system_instructions DEVELOPMENT	any	OPT-IN	The system message or instructions provided to the GenAI model separately from the chat history.	[{ "type": "text", "content": "You are an Agent that greet users, always use greetings…
gen_ai.tool.definitions DEVELOPMENT	any	OPT-IN	The list of tool definitions available to the GenAI agent or model.	[{ "type": "function", "name": "get_current_weather", "description": "Get the current…

Invoke agent internal span

SOURCE ↗

SPAN NAME
SHOULD be invoke_agent {gen_ai.agent.name} if gen_ai.agent.name is readily available. When gen_ai.agent.name is not available, it SHOULD be invoke_agent. Semantic conventions for individual GenAI systems and frameworks MAY specify different span name format.

SPAN KIND
SHOULD be INTERNAL.

ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
gen_ai.operation.name DEVELOPMENT	string	REQUIRED	The name of the operation being performed.	chat; generate_content; text_completion

☐	ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
☐	**error.type** STABLE	string	CONDITIONALLY REQUIRED If the operation ended in an error	Describes a class of error the operation ended with.	timeout; java.net.UnknownHostException; server_certificate_invalid; 500
☐	**gen_ai.agent.description** DEVELOPMENT	string	CONDITIONALLY REQUIRED When available	The free-form description of the invoked GenAI agent.	Helps with math problems; Generates fiction stories
☐	**gen_ai.agent.name** DEVELOPMENT	string	CONDITIONALLY REQUIRED When available	The human-readable name of the invoked GenAI agent.	Math Tutor; Fiction Writer
☐	**gen_ai.conversation.id** DEVELOPMENT	string	CONDITIONALLY REQUIRED	The unique identifier for a conversation (session, thread), used to store and correlate messages within this conversation.	conv_5j66UpCpwteGg4YSxUnt7lPY
☐	**gen_ai.data_source.id** DEVELOPMENT	string	CONDITIONALLY REQUIRED If applicable	The data source identifier.	H7STPQYOND
☐	**gen_ai.output.type** DEVELOPMENT	string	CONDITIONALLY REQUIRED	Represents the content type requested by the client.	text; json; image
☐	**gen_ai.request.choice.count** DEVELOPMENT	int	CONDITIONALLY REQUIRED If available, in the request, and !=1	The target number of candidate completions to return.	3
☐	**gen_ai.request.seed** DEVELOPMENT	int	CONDITIONALLY REQUIRED If applicable and if the request includes a seed	Requests with same seed value more likely to return same result.	100
☐	**gen_ai.request.frequency_penalty** DEVELOPMENT	double	RECOMMENDED	The frequency penalty setting for the GenAI request.	0.1
☐	**gen_ai.request.max_tokens** DEVELOPMENT	int	RECOMMENDED	The maximum number of tokens the model generates for a request.	100
☐	**gen_ai.request.model** DEVELOPMENT	string	RECOMMENDED If applicable	The name of the GenAI model configured for the agent.	gpt-4
☐	**gen_ai.request.presence_penalty** DEVELOPMENT	double	RECOMMENDED	The presence penalty setting for the GenAI request.	0.1
☐	**gen_ai.request.stop_sequences** DEVELOPMENT	string[]	RECOMMENDED	List of sequences that the model will use to stop generating further tokens.	["forest", "lived"]

☐	ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
☐	**gen_ai.request.temperature** DEVELOPMENT	double	RECOMMENDED	The temperature setting for the GenAI request.	0.0
☐	**gen_ai.request.top_p** DEVELOPMENT	double	RECOMMENDED	The top_p sampling setting for the GenAI request.	1.0
☐	**gen_ai.response.finish_reasons** DEVELOPMENT	string[]	RECOMMENDED	Array of reasons the model stopped generating tokens, corresponding to each generation received.	["stop"]; ["stop", "length"]; ["stop", "length", "error"]
☐	**gen_ai.usage.input_tokens** DEVELOPMENT	int	RECOMMENDED	The number of tokens used in the GenAI input (prompt).	100
☐	**gen_ai.usage.output_tokens** DEVELOPMENT	int	RECOMMENDED	The number of tokens used in the GenAI response (completion).	180
☐	**gen_ai.input.messages** DEVELOPMENT	any	OPT-IN	The chat history provided to the model as an input.	[{ "role": "user", "parts": [{ "type": "text", "content": "Weather in Paris?" }] }, {…
☐	**gen_ai.output.messages** DEVELOPMENT	any	OPT-IN	Messages returned by the model where each message represents a specific model response (choice, candidate).	[{ "role": "assistant", "parts": [{ "type": "text", "content": "The weather in Paris is…
☐	**gen_ai.system_instructions** DEVELOPMENT	any	OPT-IN	The system message or instructions provided to the GenAI model separately from the chat history.	[{ "type": "text", "content": "You are an Agent that greet users, always use greetings…
☐	**gen_ai.tool.definitions** DEVELOPMENT	any	OPT-IN	The list of tool definitions available to the GenAI agent or model.	[{ "type": "function", "name": "get_current_weather", "description": "Get the current…

Invoke workflow span

SOURCE ↗

SPAN NAME
SHOULD be invoke_workflow {gen_ai.workflow.name}.

SPAN KIND
SHOULD be INTERNAL.

ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
gen_ai.operation.name DEVELOPMENT	string	REQUIRED	The name of the operation being performed.	chat; generate_content; text_completion
error.type STABLE	string	CONDITIONALLY REQUIRED If the operation ended in an error	Describes a class of error the operation ended with.	timeout; java.net.UnknownHostException; server_certificate_invalid; 500
gen_ai.conversation.id DEVELOPMENT	string	CONDITIONALLY REQUIRED	The unique identifier for a conversation (session, thread), used to store and correlate messages within this conversation.	conv_5j66UpCpwteGg4YSxUnt7lPY
gen_ai.workflow.name DEVELOPMENT	string	CONDITIONALLY REQUIRED When available	Human-readable name of the GenAI workflow provided by the application.	multi_agent_rag; customer_support_pipeline
gen_ai.input.messages DEVELOPMENT	any	OPT-IN	The chat history provided to the model as an input.	[{ "role": "user", "parts": [{ "type": "text", "content": "Weather in Paris?" }] }, {…
gen_ai.output.messages DEVELOPMENT	any	OPT-IN	Messages returned by the model where each message represents a specific model response (choice, candidate).	[{ "role": "assistant", "parts": [{ "type": "text", "content": "The weather in Paris is…

Plan span

SOURCE ↗

SPAN NAME
SHOULD be plan {gen_ai.agent.name} if gen_ai.agent.name is readily available. When gen_ai.agent.name is not available, it SHOULD be plan.

SPAN KIND
SHOULD be INTERNAL.

ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
gen_ai.operation.name DEVELOPMENT	string	REQUIRED	The name of the operation being performed.	chat; generate_content; text_completion
error.type STABLE	string	CONDITIONALLY REQUIRED If the operation ended in an error	Describes a class of error the operation ended with.	timeout; java.net.UnknownHostException; server_certificate_invalid; 500
gen_ai.agent.name DEVELOPMENT	string	CONDITIONALLY REQUIRED When available	The human-readable name of the agent performing the planning.	Math Tutor; Fiction Writer

Events

Event names are fixed by the source headings; apply the attribute requirements for the selected event.

gen_ai.client.inference.operation.details

SOURCE ↗

ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
gen_ai.operation.name DEVELOPMENT	string	REQUIRED	The name of the operation being performed.	chat; generate_content; text_completion
gen_ai.provider.name DEVELOPMENT	string	REQUIRED	The Generative AI provider as identified by the client or server instrumentation.	openai; gcp.gen_ai; gcp.vertex_ai
error.type STABLE	string	CONDITIONALLY REQUIRED If the operation ended in an error	Describes a class of error the operation ended with.	timeout; java.net.UnknownHostException; server_certificate_invalid; 500
gen_ai.conversation.id DEVELOPMENT	string	CONDITIONALLY REQUIRED	The unique identifier for a conversation (session, thread), used to store and correlate messages within this conversation.	conv_5j66UpCpwteGg4YSxUnt7lPY
gen_ai.output.type DEVELOPMENT	string	CONDITIONALLY REQUIRED	Represents the content type requested by the client.	text; json; image
gen_ai.prompt.name DEVELOPMENT	string	CONDITIONALLY REQUIRED when a named prompt template is used	The name of the prompt that uniquely identifies it.	analyze-code
gen_ai.prompt.version DEVELOPMENT	string	CONDITIONALLY REQUIRED	The version of the prompt template used.	1.0.0; 2025-05-01; prod; v2
gen_ai.request.choice.count DEVELOPMENT	int	CONDITIONALLY REQUIRED If available, in the request, and !=1	The target number of candidate completions to return.	3
gen_ai.request.model DEVELOPMENT	string	CONDITIONALLY REQUIRED If available	The name of the GenAI model a request is being made to.	gpt-4
gen_ai.request.seed DEVELOPMENT	int	CONDITIONALLY REQUIRED If applicable and if the request includes a seed	Requests with same seed value more likely to return same result.	100

☐	ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
☐	**gen_ai.request.stream** DEVELOPMENT	boolean	CONDITIONALLY REQUIRED	Indicates whether the GenAI request was made in streaming mode.	–
☐	**gen_ai.request.top_k** DEVELOPMENT	int	CONDITIONALLY REQUIRED If applicable	The top-K sampling setting for the GenAI request: restricts token generation at each step to the K most likely next tokens.	40
☐	**server.port** STABLE	int	CONDITIONALLY REQUIRED If server.address is set	GenAI server port.	80; 8080; 443
☐	**gen_ai.conversation.compacted** DEVELOPMENT	boolean	RECOMMENDED when available	Indicates whether the effective conversation context used for this operation is a compacted view of a prior conversation.	true
☐	**gen_ai.request.frequency_penalty** DEVELOPMENT	double	RECOMMENDED	The frequency penalty setting for the GenAI request.	0.1
☐	**gen_ai.request.max_tokens** DEVELOPMENT	int	RECOMMENDED	The maximum number of tokens the model generates for a request.	100
☐	**gen_ai.request.presence_penalty** DEVELOPMENT	double	RECOMMENDED	The presence penalty setting for the GenAI request.	0.1
☐	**gen_ai.request.previous_response.id** DEVELOPMENT	string	RECOMMENDED	The unique identifier of a previous response or interaction used to provide context for the current operation.	resp_0123456789aBCdef; interaction-123
☐	**gen_ai.request.reasoning.level** DEVELOPMENT	string	RECOMMENDED When applicable	The reasoning or thinking effort level requested for a GenAI model.	low; medium; high
☐	**gen_ai.request.stop_sequences** DEVELOPMENT	string[]	RECOMMENDED	List of sequences that the model will use to stop generating further tokens.	["forest", "lived"]
☐	**gen_ai.request.temperature** DEVELOPMENT	double	RECOMMENDED	The temperature setting for the GenAI request.	0.0
☐	**gen_ai.request.top_p** DEVELOPMENT	double	RECOMMENDED	The top_p sampling setting for the GenAI request.	1.0
☐	**gen_ai.response.finish_reasons** DEVELOPMENT	string[]	RECOMMENDED	Array of reasons the model stopped generating tokens, corresponding to each generation received.	["stop"]; ["stop", "length"]; ["stop", "length", "error"]

☐	ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
☐	**gen_ai.response.id** DEVELOPMENT	string	RECOMMENDED	The unique identifier for the completion.	chatcmpl-123
☐	**gen_ai.response.model** DEVELOPMENT	string	RECOMMENDED	The name of the model that generated the response.	gpt-4-0613
☐	**gen_ai.response.time_to_first_chunk** DEVELOPMENT	double	RECOMMENDED If the request was a streaming request	Time to first chunk in a streaming response, measured from request issuance, in seconds. The value is measured from when the client issues the generation request to when…	0.5; 1.2
☐	**gen_ai.usage.audio.cache_read.input_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of audio input tokens served from a provider-managed cache.	60
☐	**gen_ai.usage.audio.input_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of audio input tokens.	120
☐	**gen_ai.usage.audio.output_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of audio output tokens.	240
☐	**gen_ai.usage.cache_read.input_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of input tokens served from a provider-managed cache.	50
☐	**gen_ai.usage.cache_write.input_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of input tokens written to a provider-managed cache.	25
☐	**gen_ai.usage.image.cache_read.input_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of image input tokens served from a provider-managed cache.	128
☐	**gen_ai.usage.image.input_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of image input tokens.	258
☐	**gen_ai.usage.image.output_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of image output tokens.	1290
☐	**gen_ai.usage.input_tokens** DEVELOPMENT	int	RECOMMENDED	The number of tokens used in the GenAI input (prompt).	100

☐	ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
☐	**gen_ai.usage.output_tokens** DEVELOPMENT	int	RECOMMENDED	The number of tokens used in the GenAI response (completion).	180
☐	**gen_ai.usage.reasoning.output_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of output tokens used for reasoning (e.g. chain-of-thought, extended thinking).	50
☐	**gen_ai.usage.text.cache_read.input_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of text input tokens served from a provider-managed cache.	40
☐	**gen_ai.usage.text.input_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of text input tokens.	100
☐	**gen_ai.usage.text.output_tokens** DEVELOPMENT	int	RECOMMENDED When applicable	The number of text output tokens.	180
☐	**server.address** STABLE	string	RECOMMENDED	GenAI server address.	example.com; 10.1.2.80; /tmp/my.sock
☐	**gen_ai.input.messages** DEVELOPMENT	any	OPT-IN	The chat history provided to the model as an input.	[{ "role": "user", "parts": [{ "type": "text", "content": "Weather in Paris?" }] }, {…
☐	**gen_ai.output.messages** DEVELOPMENT	any	OPT-IN	Messages returned by the model where each message represents a specific model response (choice, candidate).	[{ "role": "assistant", "parts": [{ "type": "text", "content": "The weather in Paris is…
☐	**gen_ai.prompt.variable** DEVELOPMENT	string	OPT-IN	The variables supplied to the prompt template, the being the variable name, the value being the variable value.	Alice; French
☐	**gen_ai.system_instructions** DEVELOPMENT	any	OPT-IN	The system message or instructions provided to the GenAI model separately from the chat history.	[{ "type": "text", "content": "You are an Agent that greet users, always use greetings…
☐	**gen_ai.tool.definitions** DEVELOPMENT	any	OPT-IN	The list of tool definitions available to the GenAI agent or model.	[{ "type": "function", "name": "get_current_weather", "description": "Get the current…

gen_ai.evaluation.result

SOURCE ↗

ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
gen_ai.evaluation.name DEVELOPMENT	string	REQUIRED	The name of the evaluation metric used for the GenAI response.	`Relevance; IntentResolution`
error.type STABLE	string	CONDITIONALLY REQUIRED If the operation ended in an error	Describes a class of error the operation ended with.	`timeout; java.net.UnknownHostException; server_certificate_invalid; 500`
gen_ai.evaluation.score.label DEVELOPMENT	string	CONDITIONALLY REQUIRED If applicable	Human readable label for evaluation.	`relevant; not_relevant; correct; incorrect; pass; fail`
gen_ai.evaluation.score.value DEVELOPMENT	double	CONDITIONALLY REQUIRED If applicable	The evaluation score returned by the evaluator.	`4.0`
gen_ai.evaluation.explanation DEVELOPMENT	string	RECOMMENDED	A free-form explanation for the assigned score provided by the evaluator.	`The response is factually accurate but lacks sufficient detail to fully address the…`
gen_ai.response.id DEVELOPMENT	string	RECOMMENDED When available	The unique identifier for the completion.	`chatcmpl-123`

MEASUREMENTS

Metrics

Each metric row preserves its published instrument type, UCUM unit, stability, and attribute set.

gen_ai.client.token.usage

SOURCE ↗

INSTRUMENT	UNIT	STABILITY	DESCRIPTION
Histogram	{token}	DEVELOPMENT	Number of input and output tokens used.

KEY ATTRIBUTE	REQUIREMENT
gen_ai.operation.name	REQUIRED
gen_ai.provider.name	REQUIRED
gen_ai.token.type	REQUIRED
gen_ai.request.model	CONDITIONALLY REQUIRED If available
server.port	CONDITIONALLY REQUIRED If server.address is set
gen_ai.response.model	RECOMMENDED
server.address	RECOMMENDED

gen_ai.client.operation.duration

SOURCE ↗

INSTRUMENT	UNIT	STABILITY	DESCRIPTION
Histogram	s	DEVELOPMENT	GenAI operation duration.

	KEY ATTRIBUTE	REQUIREMENT
☐	gen_ai.operation.name	REQUIRED
☐	error.type	CONDITIONALLY REQUIRED If the operation ended in an error
☐	gen_ai.provider.name	CONDITIONALLY REQUIRED
☐	gen_ai.request.model	CONDITIONALLY REQUIRED If available
☐	server.port	CONDITIONALLY REQUIRED If server.address is set
☐	gen_ai.response.model	RECOMMENDED
☐	server.address	RECOMMENDED

gen_ai.client.operation.time_to_first_chunk

SOURCE ↗

INSTRUMENT	UNIT	STABILITY	DESCRIPTION
Histogram	s	DEVELOPMENT	Time to receive the first chunk, measured from when the client issues the generation request to when the first chunk is received in the response stream.

	KEY ATTRIBUTE	REQUIREMENT
☐	gen_ai.operation.name	REQUIRED
☐	gen_ai.provider.name	REQUIRED
☐	gen_ai.request.model	CONDITIONALLY REQUIRED If available
☐	server.port	CONDITIONALLY REQUIRED If server.address is set
☐	gen_ai.response.model	RECOMMENDED

	KEY ATTRIBUTE	REQUIREMENT
☐	server.address	RECOMMENDED

gen_ai.client.operation.time_per_output_chunk

SOURCE ↗

INSTRUMENT	UNIT	STABILITY	DESCRIPTION
Histogram	s	DEVELOPMENT	Time per output chunk, recorded for each chunk received after the first one, measured as the time elapsed from the end of the previous chunk to the end…

	KEY ATTRIBUTE	REQUIREMENT
☐	gen_ai.operation.name	REQUIRED
☐	gen_ai.provider.name	REQUIRED
☐	gen_ai.request.model	CONDITIONALLY REQUIRED If available
☐	server.port	CONDITIONALLY REQUIRED If server.address is set
☐	gen_ai.response.model	RECOMMENDED
☐	server.address	RECOMMENDED

gen_ai.server.request.duration

SOURCE ↗

INSTRUMENT	UNIT	STABILITY	DESCRIPTION
Histogram	s	DEVELOPMENT	Generative AI server request duration such as time-to-last byte or last output token.

	KEY ATTRIBUTE	REQUIREMENT
☐	gen_ai.operation.name	REQUIRED

	KEY ATTRIBUTE	REQUIREMENT
☐	gen_ai.provider.name	REQUIRED
☐	error.type	CONDITIONALLY REQUIRED If the operation ended in an error
☐	gen_ai.request.model	CONDITIONALLY REQUIRED If available
☐	server.port	CONDITIONALLY REQUIRED If server.address is set
☐	gen_ai.response.model	RECOMMENDED
☐	server.address	RECOMMENDED

gen_ai.server.time_per_output_token

SOURCE ↗

INSTRUMENT	UNIT	STABILITY	DESCRIPTION
Histogram	s	DEVELOPMENT	Time per output token generated after the first token for successful responses.

	KEY ATTRIBUTE	REQUIREMENT
☐	gen_ai.operation.name	REQUIRED
☐	gen_ai.provider.name	REQUIRED
☐	gen_ai.request.model	CONDITIONALLY REQUIRED If available
☐	server.port	CONDITIONALLY REQUIRED If server.address is set
☐	gen_ai.response.model	RECOMMENDED
☐	server.address	RECOMMENDED

gen_ai.server.time_to_first_token

SOURCE ↗

INSTRUMENT	UNIT	STABILITY	DESCRIPTION
Histogram	s	DEVELOPMENT	Time to generate first token for successful responses.

	KEY ATTRIBUTE	REQUIREMENT
☐	gen_ai.operation.name	REQUIRED
☐	gen_ai.provider.name	REQUIRED
☐	gen_ai.request.model	CONDITIONALLY REQUIRED If available
☐	server.port	CONDITIONALLY REQUIRED If server.address is set
☐	gen_ai.response.model	RECOMMENDED
☐	server.address	RECOMMENDED

gen_ai.invoke_workflow.duration

SOURCE ↗

INSTRUMENT	UNIT	STABILITY	DESCRIPTION
Histogram	s	DEVELOPMENT	Records duration of GenAI workflow.

	KEY ATTRIBUTE	REQUIREMENT
☐	error.type	CONDITIONALLY REQUIRED If the operation ended in an error
☐	gen_ai.workflow.name	CONDITIONALLY REQUIRED If available

gen_ai.invoke_agent.duration

SOURCE ↗

INSTRUMENT	UNIT	STABILITY	DESCRIPTION
Histogram	s	DEVELOPMENT	The end-to-end duration of a single in-process agent invocation, from the moment the invocation starts until the agent emits the last chunk of its final…

KEY ATTRIBUTE	REQUIREMENT
error.type	CONDITIONALLY REQUIRED If the operation ended in an error
gen_ai.agent.name	CONDITIONALLY REQUIRED When available
gen_ai.request.model	RECOMMENDED If applicable

gen_ai.invoke_agent.inference_calls

SOURCE ↗

INSTRUMENT	UNIT	STABILITY	DESCRIPTION
Histogram	{inference_call}	DEVELOPMENT	The number of inference (model) calls a GenAI agent makes during a single invocation.

KEY ATTRIBUTE	REQUIREMENT
gen_ai.agent.name	RECOMMENDED

gen_ai.invoke_agent.tool_calls

SOURCE ↗

INSTRUMENT	UNIT	STABILITY	DESCRIPTION
Histogram	{tool_call}	DEVELOPMENT	The number of tool calls a GenAI agent makes during a single invocation.

	KEY ATTRIBUTE	REQUIREMENT
☐	gen_ai.agent.name	RECOMMENDED

gen_ai.execute_tool.duration

SOURCE ↗

INSTRUMENT	UNIT	STABILITY	DESCRIPTION
Histogram	s	DEVELOPMENT	The duration of a single tool execution.

	KEY ATTRIBUTE	REQUIREMENT
☐	gen_ai.tool.name	REQUIRED
☐	error.type	CONDITIONALLY REQUIRED If the operation ended in an error
☐	gen_ai.agent.name	CONDITIONALLY REQUIRED When applicable
☐	gen_ai.tool.type	RECOMMENDED If available

MODEL CONTEXT PROTOCOL

MCP

The MCP source defines client and server spans plus four duration metrics. Its tables also reference GenAI tool and prompt attributes where applicable.

Client

SOURCE ↗

SPAN NAME

SHOULD follow the format {mcp.method.name} {target} where target SHOULD match {gen_ai.tool.name} or {gen_ai.prompt.name} when applicable. If there is no low-cardinality target available, the Span name SHOULD be {mcp.method.name}.

SPAN KIND

SHOULD be CLIENT.

ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
mcp.method.name DEVELOPMENT	string	REQUIRED	The name of the request or notification method.	notifications/cancelled; initialize; notifications/initialized
error.type STABLE	string	CONDITIONALLY REQUIRED If and only if the operation fails	Describes a class of error the operation ended with.	timeout; java.net.UnknownHostException; server_certificate_invalid; 500
gen_ai.prompt.name DEVELOPMENT	string	CONDITIONALLY REQUIRED When operation is related to a specific prompt	The name of the prompt or prompt template provided in the request or response.	analyze-code
gen_ai.tool.name DEVELOPMENT	string	CONDITIONALLY REQUIRED When operation is related to a specific tool	Name of the tool utilized by the agent.	Flights
jsonrpc.request.id DEVELOPMENT	string	CONDITIONALLY REQUIRED When the client executes a request	A string representation of the id property of the request and its corresponding response.	10; request-7
mcp.resource.uri DEVELOPMENT	string	CONDITIONALLY REQUIRED	The value of the resource uri.	postgres://database/customers/schema; file:///home/user/documents/report.pdf
rpc.response.status_code RELEASE CANDIDATE	string	CONDITIONALLY REQUIRED If response contains an error code	The error code from the JSON-RPC response.	OK; DEADLINE_EXCEEDED; -32602
gen_ai.operation.name DEVELOPMENT	string	RECOMMENDED	The name of the GenAI operation being performed.	execute_tool

☐	ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
☐	**jsonrpc.protocol.version** DEVELOPMENT	string	RECOMMENDED When it's not 2.0	Protocol version, as specified in the jsonrpc property of the request and its corresponding response.	2.0; 1.0
☐	**mcp.protocol.version** DEVELOPMENT	string	RECOMMENDED	The version of the Model Context Protocol used.	2025-06-18
☐	**mcp.session.id** DEVELOPMENT	string	RECOMMENDED	Identifies MCP session.	191c4850af6c49e08843a3f6c80e5046
☐	**network.protocol.name** STABLE	string	RECOMMENDED When applicable	OSI application layer or non-OSI equivalent.	http; websocket
☐	**network.protocol.version** STABLE	string	RECOMMENDED When applicable	The actual version of the protocol used for network communication.	1.1; 2
☐	**network.transport** STABLE	string	RECOMMENDED	The transport protocol used for the MCP session.	tcp; quic; pipe
☐	**server.address** STABLE	string	RECOMMENDED If applicable	Server domain name if available without reverse DNS lookup; otherwise, IP address or UNIX domain socket name.	example.com; 10.1.2.80; /tmp/my.sock
☐	**server.port** STABLE	int	RECOMMENDED When server.address is set	Server port number.	80; 8080; 443
☐	**gen_ai.prompt.variable** DEVELOPMENT	string	OPT-IN	The variables supplied to the prompt template in the request.	Alice; French
☐	**gen_ai.tool.call.arguments** DEVELOPMENT	any	OPT-IN	Parameters passed to the tool call.	{ "location": "San Francisco?", "date": "2025-10-01" }
☐	**gen_ai.tool.call.result** DEVELOPMENT	any	OPT-IN	The result returned by the tool call (if any and if execution was successful).	{ "temperature_range": { "high": 75, "low": 60 }, "conditions": "sunny" }

Server

SOURCE ↗

SPAN NAME

SHOULD follow the format {mcp.method.name} {target} where target SHOULD match {gen_ai.tool.name} or {gen_ai.prompt.name} when applicable. If there is no low-cardinality target available, the Span name SHOULD be {mcp.method.name}.

SPAN KIND

SHOULD be SERVER.

ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
mcp.method.name DEVELOPMENT	string	REQUIRED	The name of the request or notification method.	notifications/cancelled; initialize; notifications/initialized
error.type STABLE	string	CONDITIONALLY REQUIRED If and only if the operation fails	Describes a class of error the operation ended with.	timeout; java.net.UnknownHostException; server_certificate_invalid; 500
gen_ai.prompt.name DEVELOPMENT	string	CONDITIONALLY REQUIRED When operation is related to a specific prompt	The name of the prompt or prompt template provided in the request or response.	analyze-code
gen_ai.tool.name DEVELOPMENT	string	CONDITIONALLY REQUIRED When operation is related to a specific tool	Name of the tool utilized by the agent.	Flights
jsonrpc.request.id DEVELOPMENT	string	CONDITIONALLY REQUIRED When the client executes a request	A string representation of the id property of the request and its corresponding response.	10; request-7
mcp.resource.uri DEVELOPMENT	string	CONDITIONALLY REQUIRED	The value of the resource uri.	postgres://database/customers/schema; file:///home/user/documents/report.pdf
rpc.response.status_code RELEASE CANDIDATE	string	CONDITIONALLY REQUIRED If response contains an error code	The error code from the JSON-RPC response.	OK; DEADLINE_EXCEEDED; -32602
client.address STABLE	string	RECOMMENDED If applicable	Client address - domain name if available without reverse DNS lookup; otherwise, IP address or UNIX domain socket name.	client.example.com; 10.1.2.80; /tmp/my.sock
client.port STABLE	int	RECOMMENDED When client.address is set	Client port number.	65123
gen_ai.operation.name DEVELOPMENT	string	RECOMMENDED	The name of the GenAI operation being performed.	execute_tool

☐	ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
☐	**jsonrpc.protocol.version** DEVELOPMENT	string	RECOMMENDED When it's not 2.0	Protocol version, as specified in the jsonrpc property of the request and its corresponding response.	2.0; 1.0
☐	**mcp.protocol.version** DEVELOPMENT	string	RECOMMENDED	The version of the Model Context Protocol used.	2025-06-18
☐	**mcp.session.id** DEVELOPMENT	string	RECOMMENDED	Identifies MCP session.	191c4850af6c49e08843a3f6c80e5046
☐	**network.protocol.name** STABLE	string	RECOMMENDED When applicable	OSI application layer or non-OSI equivalent.	http; websocket
☐	**network.protocol.version** STABLE	string	RECOMMENDED When applicable	The actual version of the protocol used for network communication.	1.1; 2
☐	**network.transport** STABLE	string	RECOMMENDED	The transport protocol used for the MCP session.	tcp; quic; pipe
☐	**gen_ai.prompt.variable** DEVELOPMENT	string	OPT-IN	The variables supplied to the prompt template in the request.	Alice; French
☐	**gen_ai.tool.call.arguments** DEVELOPMENT	any	OPT-IN	Parameters passed to the tool call.	{ "location": "San Francisco?", "date": "2025-10-01" }
☐	**gen_ai.tool.call.result** DEVELOPMENT	any	OPT-IN	The result returned by the tool call (if any and if execution was successful).	{ "temperature_range": { "high": 75, "low": 60 }, "conditions": "sunny" }

MCP metrics

mcp.client.operation.duration

SOURCE ↗

INSTRUMENT	UNIT	STABILITY	DESCRIPTION
Histogram	s	DEVELOPMENT	The duration of the MCP request or notification as observed on the sender from the time it was sent until the response or ack is received.

	KEY ATTRIBUTE	REQUIREMENT
☐	mcp.method.name	REQUIRED
☐	error.type	CONDITIONALLY REQUIRED If and only if the operation fails
☐	gen_ai.prompt.name	CONDITIONALLY REQUIRED When operation is related to a specific prompt
☐	gen_ai.tool.name	CONDITIONALLY REQUIRED When operation is related to a specific tool
☐	rpc.response.status_code	CONDITIONALLY REQUIRED If response contains an error code
☐	gen_ai.operation.name	RECOMMENDED
☐	jsonrpc.protocol.version	RECOMMENDED When it's not 2.0
☐	mcp.protocol.version	RECOMMENDED
☐	network.protocol.name	RECOMMENDED When applicable
☐	network.protocol.version	RECOMMENDED When applicable
☐	network.transport	RECOMMENDED
☐	server.address	RECOMMENDED If applicable
☐	server.port	RECOMMENDED When server.address is set
☐	gen_ai.prompt.variable	OPT-IN
☐	mcp.resource.uri	OPT-IN

mcp.server.operation.duration

SOURCE ↗

INSTRUMENT	UNIT	STABILITY	DESCRIPTION
Histogram	s	DEVELOPMENT	MCP request or notification duration as observed on the receiver from the time it was received until the result or ack is sent.

KEY ATTRIBUTE	REQUIREMENT
mcp.method.name	REQUIRED
error.type	CONDITIONALLY REQUIRED If and only if the operation fails
gen_ai.prompt.name	CONDITIONALLY REQUIRED When operation is related to a specific prompt
gen_ai.tool.name	CONDITIONALLY REQUIRED When operation is related to a specific tool
rpc.response.status_code	CONDITIONALLY REQUIRED If response contains an error code
gen_ai.operation.name	RECOMMENDED
jsonrpc.protocol.version	RECOMMENDED When it's not 2.0
mcp.protocol.version	RECOMMENDED
network.protocol.name	RECOMMENDED When applicable
network.protocol.version	RECOMMENDED When applicable
network.transport	RECOMMENDED
gen_ai.prompt.variable	OPT-IN
mcp.resource.uri	OPT-IN

mcp.client.session.duration

SOURCE ↗

INSTRUMENT	UNIT	STABILITY	DESCRIPTION
Histogram	s	DEVELOPMENT	The duration of the MCP session as observed on the MCP client.

KEY ATTRIBUTE	REQUIREMENT
error.type	CONDITIONALLY REQUIRED If and only if session ends with an error
jsonrpc.protocol.version	RECOMMENDED When it's not 2.0
mcp.protocol.version	RECOMMENDED
network.protocol.name	RECOMMENDED When applicable
network.protocol.version	RECOMMENDED When applicable
network.transport	RECOMMENDED
server.address	RECOMMENDED If applicable
server.port	RECOMMENDED When server.address is set

mcp.server.session.duration

SOURCE ↗

INSTRUMENT	UNIT	STABILITY	DESCRIPTION
Histogram	s	DEVELOPMENT	The duration of the MCP session as observed on the MCP server.

	KEY ATTRIBUTE	REQUIREMENT
☐	`error.type`	CONDITIONALLY REQUIRED If and only if session ends with an error
☐	`jsonrpc.protocol.version`	RECOMMENDED When it's not 2.0
☐	`mcp.protocol.version`	RECOMMENDED
☐	`network.protocol.name`	RECOMMENDED When applicable
☐	`network.protocol.version`	RECOMMENDED When applicable
☐	`network.transport`	RECOMMENDED

Provider-specific additions

This section lists only provider-page attribute keys absent from the generic GenAI signal pages. Provider span conventions are retained even when no unique key is added.

OpenAI

SOURCE ↗

SPAN NAME

SHOULD be {gen_ai.operation.name} {gen_ai.request.model}. Semantic conventions for individual GenAI systems and frameworks MAY specify different span name format and MUST follow the overall guidelines for span names.

SPAN KIND

SHOULD be CLIENT.

Only keys present in this provider page and absent from the generic GenAI signal pages are listed.

☐	ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
☐	**openai.request.service_tier** DEVELOPMENT	string	CONDITIONALLY REQUIRED	The service tier requested. May be a specific tier, default, or auto.	auto; default
☐	**openai.response.service_tier** DEVELOPMENT	string	CONDITIONALLY REQUIRED	The service tier used for the response.	scale; default
☐	**openai.api.type** DEVELOPMENT	string	RECOMMENDED	The type of OpenAI API being used.	chat_completions; responses
☐	**openai.response.system_fingerprint** DEVELOPMENT	string	RECOMMENDED	A fingerprint to track any eventual change in the Generative AI environment.	fp_44709d6fcb

Anthropic

SOURCE ↗

SPAN NAME

SHOULD be {gen_ai.operation.name} {gen_ai.request.model}. Semantic conventions for individual GenAI systems and frameworks MAY specify different span name format and MUST follow the overall guidelines for span names.

SPAN KIND

SHOULD be CLIENT.

No additional attribute key is unique to this provider page; apply its published mappings to the generic convention.

AWS Bedrock

SOURCE ↗

SPAN NAME

SHOULD be {gen_ai.operation.name} {gen_ai.request.model}. Semantic conventions for individual GenAI systems and frameworks MAY specify different span name format and MUST follow the overall guidelines for span names.

SPAN KIND

SHOULD be CLIENT.

Only keys present in this provider page and absent from the generic GenAI signal pages are listed.

ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
aws.bedrock.guardrail.id DEVELOPMENT	string	REQUIRED	The unique identifier of the AWS Bedrock Guardrail. A guardrail helps safeguard and prevent unwanted behavior from model responses or user messages.	sgi5gkybzqak
aws.bedrock.knowledge_base.id DEVELOPMENT	string	RECOMMENDED	The unique identifier of the AWS Bedrock Knowledge base. A knowledge base is a bank of information that can be queried by models to generate more relevant responses and…	XFWUPB9PAW

Azure AI Inference

SOURCE ↗

SPAN NAME

SHOULD be {gen_ai.operation.name} {gen_ai.request.model}. Semantic conventions for individual GenAI systems and frameworks MAY specify different span name format and MUST follow the overall guidelines for span names.

SPAN KIND

SHOULD be CLIENT.

Only keys present in this provider page and absent from the generic GenAI signal pages are listed.

ATTRIBUTE	TYPE	REQUIREMENT LEVEL	SHORT DESCRIPTION	EXAMPLE
azure.resource_provider.namespace DEVELOPMENT	string	RECOMMENDED	Azure Resource Provider Namespace as recognized by the client.	Microsoft.CognitiveServices

COMPLETE REGISTRY

Appendix A · All gen_ai.* attributes

72 current attributes, alphabetized from the pinned GenAI repository registry. The 10 truly deprecated names from the old registry are intentionally excluded here and listed in “Deprecated - do not use.”

☐	ATTRIBUTE	TYPE	STABILITY	DESCRIPTION	EXAMPLE / DEPRECATION
☐	**gen_ai.agent.description**	string	DEVELOPMENT	Free-form description of the GenAI agent provided by the application.	Helps with math problems; Generates fiction stories
☐	**gen_ai.agent.id**	string	DEVELOPMENT	The unique and stable identifier of the GenAI hosted agent resource.	asst_5j66UpCpwteGg4YSxUnt7lPY; arn:aws:bedrock:us-east-1:123:agent/42…
☐	**gen_ai.agent.name**	string	DEVELOPMENT	Human-readable name of the GenAI agent provided by the application.	Math Tutor; Fiction Writer
☐	**gen_ai.agent.version**	string	DEVELOPMENT	The version of the GenAI agent.	1.0.0; 2025-05-01
☐	**gen_ai.conversation.compacted**	boolean	DEVELOPMENT	Indicates whether the effective conversation context used for this operation is a compacted view of a prior conversation.	true
☐	**gen_ai.conversation.id**	string	DEVELOPMENT	The unique identifier for a conversation (session, thread), used to store and correlate messages within this conversation.	conv_5j66UpCpwteGg4YSxUnt7lPY
☐	**gen_ai.data_source.id**	string	DEVELOPMENT	The data source identifier.	H7STPQYOND
☐	**gen_ai.embeddings.dimension.count**	int	DEVELOPMENT	The number of dimensions the resulting output embeddings should have.	512; 1024
☐	**gen_ai.evaluation.explanation**	string	DEVELOPMENT	A free-form explanation for the assigned score provided by the evaluator.	The response is factually accurate but lacks sufficient detail to fully address the…
☐	**gen_ai.evaluation.name**	string	DEVELOPMENT	The name of the evaluation metric used for the GenAI response.	Relevance; IntentResolution
☐	**gen_ai.evaluation.score.label**	string	DEVELOPMENT	Human readable label for evaluation.	relevant; not_relevant; correct; incorrect; pass; fail
☐	**gen_ai.evaluation.score.value**	double	DEVELOPMENT	The evaluation score returned by the evaluator.	4.0
☐	**gen_ai.input.messages**	any	DEVELOPMENT	The chat history provided to the model as an input.	[{ "role": "user", "parts": [{ "type": "text", "content": "Weather in Paris?" }] }, {…

☐	ATTRIBUTE	TYPE	STABILITY	DESCRIPTION	EXAMPLE / DEPRECATION
☐	**gen_ai.memory.query.text**	string	DEVELOPMENT	The search query used to retrieve memories.	user dietary preferences; past flight bookings
☐	**gen_ai.memory.record.count**	int	DEVELOPMENT	The number of memory records relevant to the operation.	3
☐	**gen_ai.memory.record.id**	string	DEVELOPMENT	The unique identifier of the memory record.	mem_5j66UpCpwteGg4YSxUnt7lPY
☐	**gen_ai.memory.records**	any	DEVELOPMENT	The memory records stored or retrieved in a memory operation.	[{ "content": "User prefers dark mode", "id": "mem_123", "score": 0.95 }, { "content": {…
☐	**gen_ai.memory.store.id**	string	DEVELOPMENT	The unique identifier of the memory store.	ms_abc123; user-preferences-store
☐	**gen_ai.operation.name**	string	DEVELOPMENT	The name of the operation being performed.	chat; generate_content; text_completion
☐	**gen_ai.output.messages**	any	DEVELOPMENT	Messages returned by the model where each message represents a specific model response (choice, candidate).	[{ "role": "assistant", "parts": [{ "type": "text", "content": "The weather in Paris is…
☐	**gen_ai.output.type**	string	DEVELOPMENT	Represents the content type requested by the client.	text; json; image
☐	**gen_ai.prompt.name**	string	DEVELOPMENT	The name of the prompt that uniquely identifies it.	analyze-code
☐	**gen_ai.prompt.variable**	string	DEVELOPMENT	The variables supplied to the prompt template, the being the variable name, the value being the variable value.	Alice; French
☐	**gen_ai.prompt.version**	string	DEVELOPMENT	The version of the prompt template used.	1.0.0; 2025-05-01; prod; v2
☐	**gen_ai.provider.name**	string	DEVELOPMENT	The Generative AI provider as identified by the client or server instrumentation.	openai; gcp.gen_ai; gcp.vertex_ai
☐	**gen_ai.request.choice.count**	int	DEVELOPMENT	The target number of candidate completions to return.	3
☐	**gen_ai.request.encoding_formats**	string[]	DEVELOPMENT	The encoding formats requested in an embeddings operation, if specified.	["base64"]; ["float", "binary"]
☐	**gen_ai.request.frequency_penalty**	double	DEVELOPMENT	The frequency penalty setting for the GenAI request.	0.1
☐	**gen_ai.request.max_tokens**	int	DEVELOPMENT	The maximum number of tokens the model generates for a request.	100
☐	**gen_ai.request.model**	string	DEVELOPMENT	The name of the GenAI model a request is being made to.	gpt-4

☐	ATTRIBUTE	TYPE	STABILITY	DESCRIPTION	EXAMPLE / DEPRECATION
☐	**gen_ai.request.presence_penalty**	double	DEVELOPMENT	The presence penalty setting for the GenAI request.	0.1
☐	**gen_ai.request.previous_response.id**	string	DEVELOPMENT	The unique identifier of a previous response or interaction used to provide context for the current operation.	resp_0123456789aBCdef; interaction-123
☐	**gen_ai.request.reasoning.level**	string	DEVELOPMENT	The reasoning or thinking effort level requested for a GenAI model.	low; medium; high
☐	**gen_ai.request.seed**	int	DEVELOPMENT	Requests with same seed value more likely to return same result.	100
☐	**gen_ai.request.stop_sequences**	string[]	DEVELOPMENT	List of sequences that the model will use to stop generating further tokens.	["forest", "lived"]
☐	**gen_ai.request.stream**	boolean	DEVELOPMENT	Indicates whether the GenAI request was made in streaming mode.	–
☐	**gen_ai.request.stream_cursor**	string	DEVELOPMENT	The cursor identifying the last streamed event already received, used to resume a streamed response from that position.	42; event-abc123
☐	**gen_ai.request.temperature**	double	DEVELOPMENT	The temperature setting for the GenAI request.	0.0
☐	**gen_ai.request.top_k**	int	DEVELOPMENT	The top-K sampling setting for the GenAI request: restricts token generation at each step to the K most likely next tokens.	40
☐	**gen_ai.request.top_p**	double	DEVELOPMENT	The top_p sampling setting for the GenAI request.	1.0
☐	**gen_ai.response.finish_reasons**	string[]	DEVELOPMENT	Array of reasons the model stopped generating tokens, corresponding to each generation received.	["stop"]; ["stop", "length"]; ["stop", "length", "error"]
☐	**gen_ai.response.id**	string	DEVELOPMENT	The unique identifier for the completion.	chatcmpl-123
☐	**gen_ai.response.model**	string	DEVELOPMENT	The name of the model that generated the response.	gpt-4-0613
☐	**gen_ai.response.status**	string	DEVELOPMENT	The lifecycle status of a generated response, as reported by the provider when the response is fetched or polled.	completed; in_progress
☐	**gen_ai.response.time_to_first_chunk**	double	DEVELOPMENT	Time to first chunk in a streaming response, measured from request issuance, in seconds. The value is measured from when the client issues the generation request to when the first chunk is received in…	0.5; 1.2
☐	**gen_ai.retrieval.documents**	any	DEVELOPMENT	The documents retrieved.	[{ "id": "doc_123", "score": 0.95 }, { "id": "doc_456", "score": 0.87 }, { "id"…
☐	**gen_ai.retrieval.query.text**	string	DEVELOPMENT	The query text used for retrieval.	What is the capital of France?; weather in Paris

☐	ATTRIBUTE	TYPE	STABILITY	DESCRIPTION	EXAMPLE / DEPRECATION
☐	**gen_ai.retrieval.top_k**	int	DEVELOPMENT	The maximum number of documents the retriever was asked to return for the query (also known as k, limit, or max_num_results).	5
☐	**gen_ai.system_instructions**	any	DEVELOPMENT	The system message or instructions provided to the GenAI model separately from the chat history.	[{ "type": "text", "content": "You are an Agent that greet users, always use greetings…
☐	**gen_ai.token.type**	string	DEVELOPMENT	The type of token being counted.	input; output
☐	**gen_ai.tool.call.arguments**	any	DEVELOPMENT	Parameters passed to the tool call.	{ "location": "San Francisco?", "date": "2025-10-01" }
☐	**gen_ai.tool.call.id**	string	DEVELOPMENT	The tool call identifier.	call_mszuSIzqtI65i1wAUOE8w5H4
☐	**gen_ai.tool.call.result**	any	DEVELOPMENT	The result returned by the tool call (if any and if execution was successful).	{ "temperature_range": { "high": 75, "low": 60 }, "conditions": "sunny" }
☐	**gen_ai.tool.definitions**	any	DEVELOPMENT	The list of tool definitions available to the GenAI agent or model.	[{ "type": "function", "name": "get_current_weather", "description": "Get the current…
☐	**gen_ai.tool.description**	string	DEVELOPMENT	The tool description.	Multiply two numbers
☐	**gen_ai.tool.name**	string	DEVELOPMENT	Name of the tool utilized by the agent.	Flights
☐	**gen_ai.tool.type**	string	DEVELOPMENT	Type of the tool utilized by the agent	function; extension; datastore
☐	**gen_ai.usage.audio.cache_read.input_tokens**	int	DEVELOPMENT	The number of audio input tokens served from a provider-managed cache.	60
☐	**gen_ai.usage.audio.input_tokens**	int	DEVELOPMENT	The number of audio input tokens.	120
☐	**gen_ai.usage.audio.output_tokens**	int	DEVELOPMENT	The number of audio output tokens.	240
☐	**gen_ai.usage.cache_read.input_tokens**	int	DEVELOPMENT	The number of input tokens served from a provider-managed cache.	50
☐	**gen_ai.usage.cache_write.input_tokens**	int	DEVELOPMENT	The number of input tokens written to a provider-managed cache.	25
☐	**gen_ai.usage.image.cache_read.input_tokens**	int	DEVELOPMENT	The number of image input tokens served from a provider-managed cache.	128

☐	ATTRIBUTE	TYPE	STABILITY	DESCRIPTION	EXAMPLE / DEPRECATION
☐	**gen_ai.usage.image.input_tokens**	int	DEVELOPMENT	The number of image input tokens.	258
☐	**gen_ai.usage.image.output_tokens**	int	DEVELOPMENT	The number of image output tokens.	1290
☐	**gen_ai.usage.input_tokens**	int	DEVELOPMENT	The number of tokens used in the GenAI input (prompt).	100
☐	**gen_ai.usage.output_tokens**	int	DEVELOPMENT	The number of tokens used in the GenAI response (completion).	180
☐	**gen_ai.usage.reasoning.output_tokens**	int	DEVELOPMENT	The number of output tokens used for reasoning (e.g. chain-of-thought, extended thinking).	50
☐	**gen_ai.usage.text.cache_read.input_tokens**	int	DEVELOPMENT	The number of text input tokens served from a provider-managed cache.	40
☐	**gen_ai.usage.text.input_tokens**	int	DEVELOPMENT	The number of text input tokens.	100
☐	**gen_ai.usage.text.output_tokens**	int	DEVELOPMENT	The number of text output tokens.	180
☐	**gen_ai.workflow.name**	string	DEVELOPMENT	Human-readable name of the GenAI workflow provided by the application.	multi_agent_rag; customer_support_pipeline

TRACEABILITY

Appendix B · Source file mapping

Every link is pinned: the GenAI repository at `c88d504ab3d9` and the earlier semantic-conventions repository at `d0472f4ae331`. No moving-branch URL supplies checklist data.

REPOSITORY	SOURCE FILE	USED FOR
open-telemetry/semantic-conventions-genai	LICENSE	Cover: Apache License 2.0 source notice
open-telemetry/semantic-conventions-genai	docs/registry/attributes/gen-ai.md	Appendix A: complete gen_ai.* registry
open-telemetry/semantic-conventions-genai	docs/registry/attributes/mcp.md	MCP registry cross-check
open-telemetry/semantic-conventions-genai	docs/registry/attributes/openai.md	OpenAI additions cross-check
open-telemetry/semantic-conventions-genai	docs/registry/attributes/aws.md	AWS Bedrock additions cross-check
open-telemetry/semantic-conventions-genai	docs/gen-ai/gen-ai-spans.md	LLM inference, embeddings, tool execution
open-telemetry/semantic-conventions-genai	docs/gen-ai/gen-ai-agent-spans.md	Agent and workflow spans
open-telemetry/semantic-conventions-genai	docs/gen-ai/gen-ai-events.md	Events
open-telemetry/semantic-conventions-genai	docs/gen-ai/gen-ai-metrics.md	Metrics
open-telemetry/semantic-conventions-genai	docs/gen-ai/mcp.md	MCP spans and metrics
open-telemetry/semantic-conventions-genai	docs/gen-ai/openai.md	OpenAI provider additions
open-telemetry/semantic-conventions-genai	docs/gen-ai/anthropic.md	Anthropic provider mapping
open-telemetry/semantic-conventions-genai	docs/gen-ai/aws-bedrock.md	AWS Bedrock provider additions
open-telemetry/semantic-conventions-genai	docs/gen-ai/azure-ai-inference.md	Azure AI Inference provider additions
open-telemetry/semantic-conventions-genai	docs/gen-ai/README.md	Repository section index; build evidence
open-telemetry/semantic-conventions-genai	docs/gen-ai/gen-ai-exceptions.md	Fetched reference; not separately tabulated
open-telemetry/semantic-conventions-genai	docs/gen-ai/non-normative/examples-llm-calls.md	Fetched reference; examples not copied into checklist rows
open-telemetry/semantic-conventions	docs/registry/attributes/gen-ai.md	Deprecated warning table; Moved section excluded